U.S. Midterm Policy Topics in Guardian News and Their Market Associations

Topic Modeling, Democratic Sweep Prediction-Market Tests, and Equity Evidence

Kuanhong Cai

New York University

October 2026

github.com/kc6534-JerryCai/NLP_asignment5

ABSTRACT

This report studies substantive policy coverage surrounding the 2026 U.S. midterm elections in 1,216 Guardian articles collected from July 10 through October 9, 2026. A policy-focused screening stage retained 227 articles, of which 226 passed text-length requirements. A six-topic LDA model identifies technology governance, infrastructure, trade and rights, Iran and energy security, immigration and household economics, and defense. DeepSeek-based article annotations separate topic sentiment from Democratic and Republican stance. The resulting time series are compared with Polymarket's joint Democratic Senate-and-House sweep contract and with market-adjusted U.S. equity returns. The Democratic Sweep tests produce no association that survives multiplicity adjustment. Equity regressions identify broad statistically significant relationships across individual stocks and topic-specific baskets, with economically meaningful effect sizes in several policy signals. The results indicate that the policy-news measures contain market-relevant information during the study window, while the reported associations are interpreted without causal claims.

Keywords: topic modeling; LDA; news sentiment; U.S. midterms; Polymarket; market-adjusted returns; multiple testing

Principal finding	The topic model produces interpretable policy clusters and useful descriptive time series, while the market evidence differs by outcome. Democratic Sweep correlations are not statistically reliable after BH adjustment. In contrast, the equity results show broad and economically meaningful associations at both the individual-stock and dedicated-basket levels.

1. Research Design and Data

The project separates three quantities that are often conflated: *attention* (how much an article loads on a topic), *topic sentiment* (the evaluative tone toward the policy condition or proposal), and *party stance* (support for or criticism of Democrats or Republicans within the topic). Attention is obtained from LDA probabilities; sentiment and stance are article-level language-model annotations. Missing expression is never recoded as neutral, and missing news is never filled with zero.

Table 1. Core datasets

Component	Scale	Definition
Guardian corpus	1,216 English-language articles	2026-07-10 to 2026-10-09 (New York dates)
Policy LDA	226 valid policy documents	One full article body per document; K=6
Article annotations	1,356 article-topic rows	226 articles x 6 topics; no failed articles
Democratic Sweep	2,205 hourly observations	92 daily closing proxies; 14 appended out-of-range records removed
Equities	999 downloaded / 998 testable	Market-adjusted open-to-close return versus SPY
Research window	92 natural days	Calendar alignment preserves weekends and missingness

1.1 Policy-scope sample

The original screening classified 641 articles as broadly relevant, 424 as uncertain, and 151 as irrelevant. A second policy-centrality screen produced 227 policy-substance articles, 103 mixed policy/campaign articles, 249 campaign-process articles, 486 uncertain articles, and 151 excluded articles. These five categories exhaust the 1,216-article corpus. The main model intentionally uses only policy-substance articles; one document fell below the 30-token threshold, leaving 226 LDA documents. Mixed and uncertain records remain available for sensitivity analysis but are not folded into the headline model.

Interpretation boundary	All attention and sentiment estimates describe the Guardian policy-news sample. They are not measures of U.S. voter preferences, approval, voting intention, or causal media effects.

1.2 Market and equity outcomes

The prediction-market outcome is the verified Polymarket contract *2026 Balance of Power: D Senate, D House* (group label: Democrats Sweep; market ID 562828). The analysis uses the Yes token, downloaded in at-most-seven-day blocks at hourly fidelity. Each New York day is represented by its last valid observation; no cross-day forward fill is used. Equity outcomes are split-adjusted open-to-close stock returns minus the contemporaneous SPY return, which targets stock-specific movement rather than broad-market direction.

2. NLP Method: Policy-Focused Topic Modeling

2.1 Text preparation

Each complete article body is one document; titles are retained for interpretation but are not appended to the body. The pipeline removes HTML, URLs, abnormal whitespace, and source-template fragments while preserving the original article separately. spaCy (*en_core_web_sm*) lowercases, tokenizes, and lemmatizes English text. Stopwords, pure punctuation, pure numbers, and clearly non-informative tokens are removed. Gensim's phrase model detects recurring bigrams such as *service_member* and *strait_hormuz*. The acronym *AI* is explicitly retained. Election-mechanics and high-frequency political context terms are removed only from LDA tokens after inspection showed that they merged otherwise distinct policy clusters; the source text remains unchanged.

Element	Implementation
Document unit	One complete article body
Minimum usable length	30 processed tokens; 1 article excluded
Phrase model	min_count=4; threshold=8.0
Dictionary filter	no_below=4; no_above=0.75
Vocabulary	10,943 terms before filtering; 3,513 after filtering
LDA input	Gensim Dictionary + doc2bow term frequencies; no TF-IDF
Estimator	Gensim LdaModel; passes=10; iterations=100; alpha=auto; eta=auto

2.2 Candidate models and topic-number selection

Candidate models K=4 through K=8 were trained with random state 42. Selection considered c_v coherence, cross-topic keyword overlap, representative articles, topic size, and interpretability; coherence alone did not determine K. The K=6 model has the highest coherence in the candidate set and preserves a distinct AI topic while separating energy-security, defense, infrastructure, immigration/economics, and trade/rights coverage. Stability checks refit K=6 with seeds 7 and 99 and matched topics by their word distributions rather than topic IDs.

Table 2. LDA candidate comparison

K	c_v	Unique top-15	Mean overlap	Mean dominant p	Decision
4	0.367	0.967	0.011	0.867	Candidate
5	0.370	0.920	0.021	0.859	Candidate
6	0.407	0.867	0.029	0.846	Selected
7	0.390	0.829	0.033	0.834	Candidate
8	0.389	0.842	0.027	0.814	Candidate

The selected model's matched-seed mean full-vocabulary cosine similarity is 0.728 and its mean top-15 Jaccard overlap is 0.232. These values support broad reproducibility but also show that exact word rankings are not invariant. Topic names are therefore documented human interpretations, not facts emitted directly by LDA.

3. Topic Model Results

Table 3. Final policy topics and conditional attention shares

Topic	Provisional interpretation	Share	Leading terms
T0	AI, Datacenters, and Digital Governance	17.9%	ai, datacenter, china, company, public, family, technology, center
T1	Infrastructure, Public Lands, and Federal Investment	10.9%	construction, federal, city, plan, agency, job, cbp, big_bend
T2	Trade, Public Health, and Legal Rights	23.8%	tariff, canada, ukraine, trade, vaccine, court, law, russia
T3	Iran War, Oil, and Energy Security	23.6%	iran, war, attack, iranian, israel, military, strike, tehran
T4	Immigration, Public Funding, and Household Economy	12.4%	child, investment, immigration, ice, company, policy, fund, rate
T5	Defense Spending and Military Security	11.3%	war, iran, hegseth, pentagon, military, service_member, kill, city

T2 and T3 jointly account for 47.4% of probability-weighted attention. T2 is the least internally coherent topic because it combines trade, public-health, and legal-rights material; T4 is also broad, combining immigration enforcement with public funding and household economics. These mixed clusters should be interpreted with their subissue annotations rather than as single narrow policy concepts.

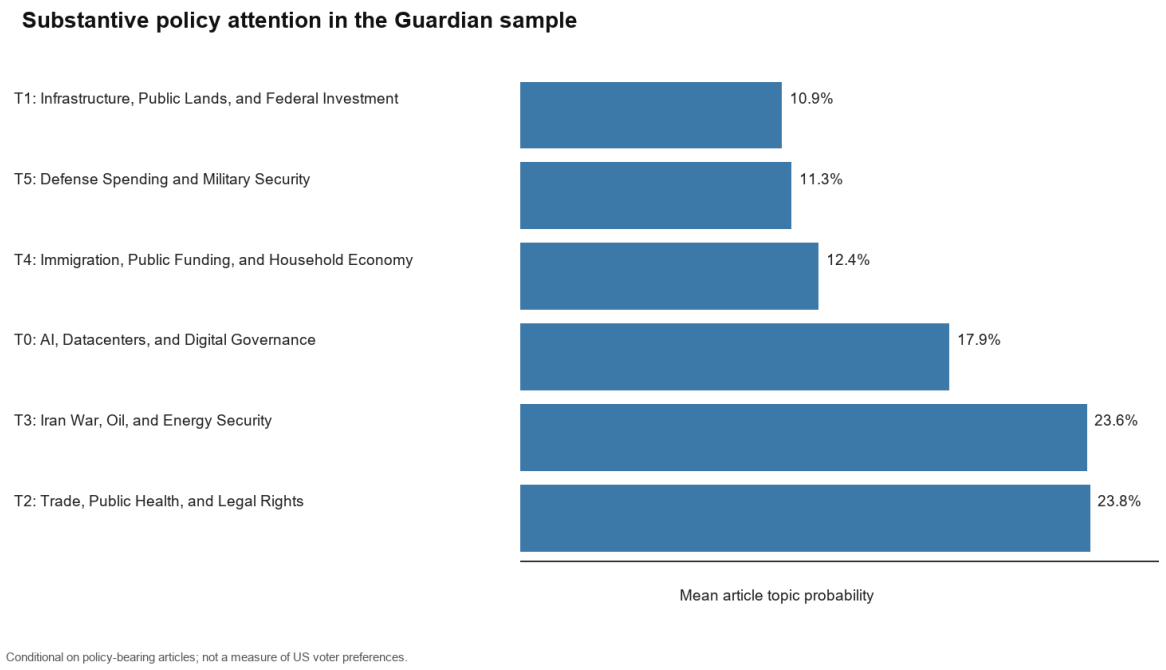


Figure 1. Overall topic attention in the 226-document policy sample. Shares are averages of complete document-topic probability vectors and sum to one; they are conditional on inclusion in the policy sample.

Why energy and AI appear

The policy-only design removes campaign-process language that dominated the earlier all-discourse model. This allows T0 (AI and digital governance) and T3 (Iran, oil, and energy security) to emerge as distinct clusters rather than being absorbed by generic election terminology.

4. Policy Attention and Sentiment over Time

Article-level sentiment is labeled positive (+1), negative (-1), neutral (0), mixed (0), or unclear/missing. Topic relevance is confirmed independently of LDA probability. Daily sentiment is the LDA-probability-weighted mean among relevant, non-missing article-topic scores. Because 226 articles are spread over 92 days, raw daily series are sparse. The figures below therefore use trailing seven-day pooled windows: all qualifying articles in each window are re-aggregated directly, rather than averaging daily means. Missing windows remain missing and are not connected through fabricated zeros.

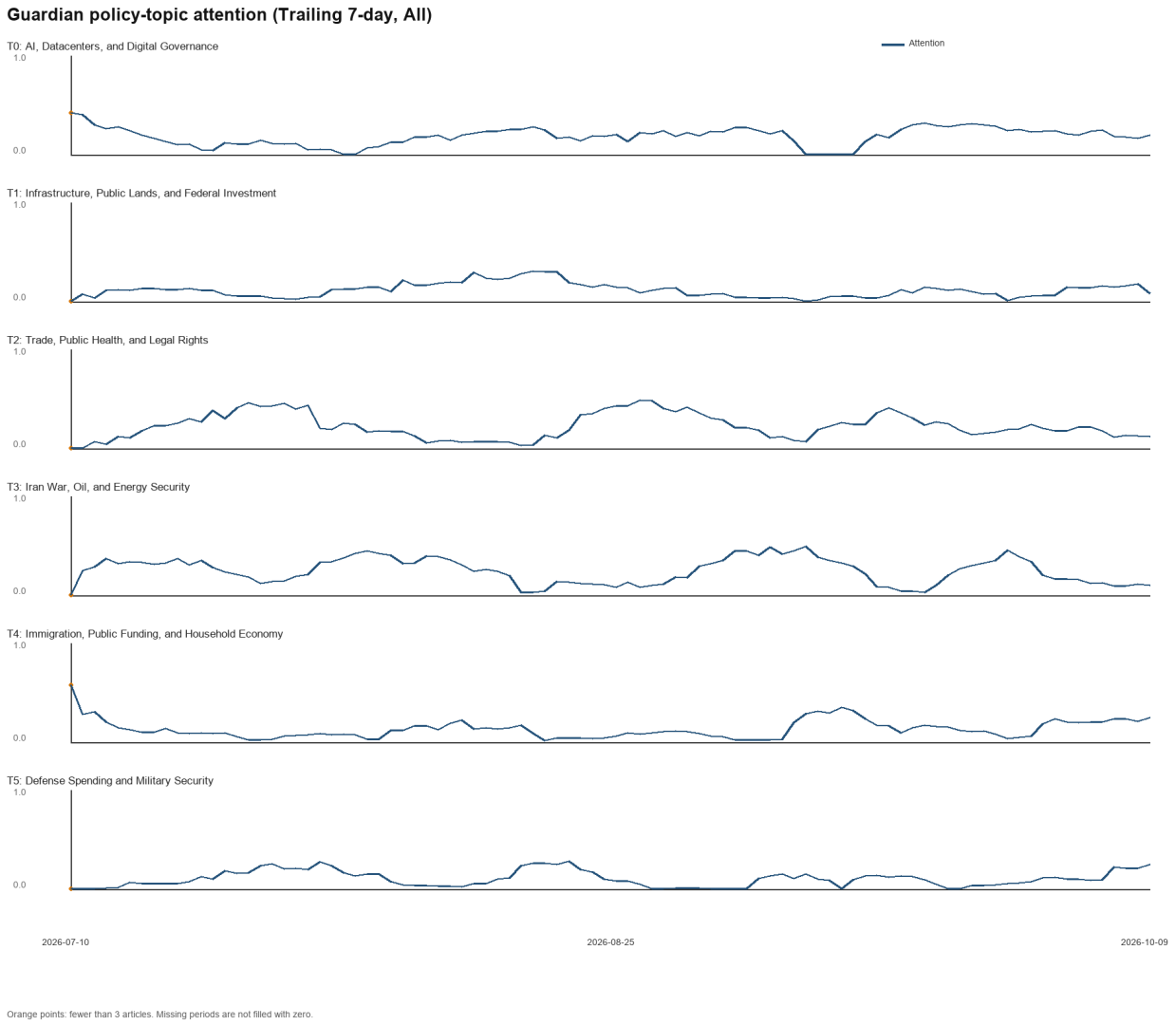


Figure 2. Trailing seven-day Guardian policy-topic attention. This is the most continuous descriptive series because every valid LDA article contributes its complete probability vector. The measure captures coverage allocation, not sentiment.

4. Policy Attention and Sentiment over Time (continued)

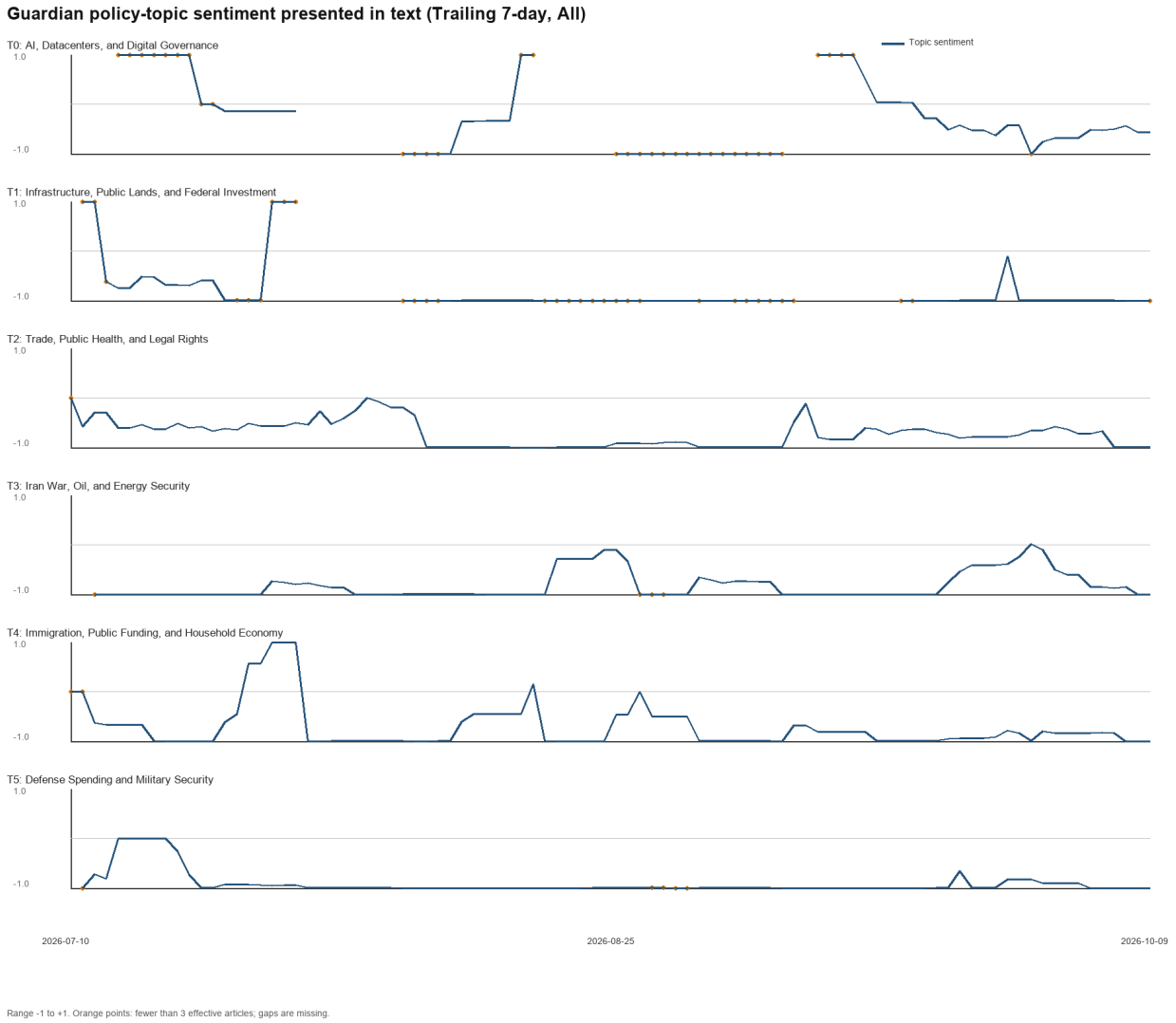


Figure 3. Trailing seven-day topic sentiment presented in Guardian policy texts. Pooling produces a substantially more continuous display than the daily chart while preserving genuine missing windows, especially for the lower-coverage T0 and T1 series. Orange markers identify low effective sample size.

Table 4. Annotation coverage by topic

Topic	Short name	Sentiment n	Article cov.	Weight cov.	Dem. n	Rep. n
T0	AI / digital governance	34	15.0%	44.9%	13	16
T1	Infrastructure / federal investment	34	15.0%	55.6%	16	20
T2	Trade / health / legal rights	159	70.4%	92.6%	68	107
T3	Iran / oil / energy security	114	50.4%	91.5%	23	57
T4	Immigration / household economy	122	54.0%	83.6%	42	68
T5	Defense / military security	93	41.2%	93.1%	31	52

The annotation stage processed all 226 articles (1,356 article-topic records), with no failed articles and no evidence strings that could not be located in the source text. Nevertheless, the label distribution is highly asymmetric: topic sentiment is predominantly negative, Democratic opposition is absent, and Republican opposition dominates the non-neutral Republican labels. This may reflect the sample period, Guardian framing, and model behavior; it is not a validated measure of electoral opinion.

5. Relationship with the Democratic Sweep Market

The market series is the Polymarket Yes price for Democrats controlling both chambers after the 2026 midterms. Hourly observations are converted from UTC to America/New_York, validated to lie in [0,1], filtered to each requested segment, sorted, and deduplicated. Fourteen API records appended outside their requested historical segments were removed. All 92 daily closing proxies pass the three-hour staleness rule. Prices are market-implied probability proxies, not objective probabilities.

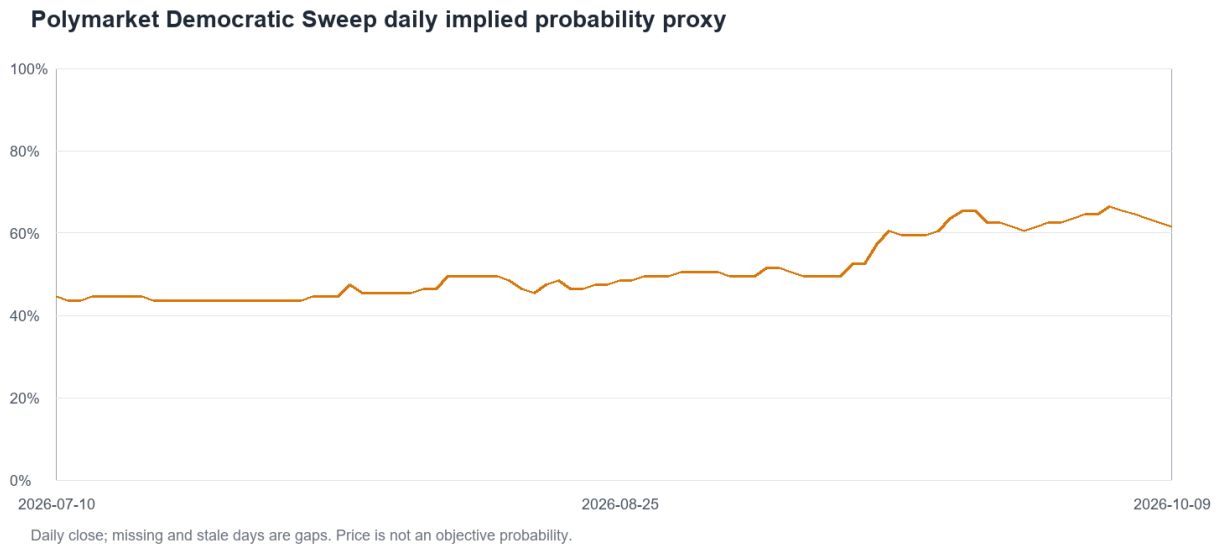


Figure 4. Democratic Sweep daily closing price proxy, July 10-October 9, 2026. The contract is the joint Senate-and-House event; separate chamber probabilities are neither substituted nor multiplied.

Table 5. Primary same-day change tests: corr(dS, dP)

Index	Topic	Pearson r	N	7-day block 95% CI	BH q	q<.05
Topic sentiment	T0	-0.316	8	[-0.918, 0.218]	0.758	No
Topic sentiment	T1	0.258	13	[-0.423, 0.686]	0.739	No
Topic sentiment	T2	0.170	61	[0.034, 0.352]	0.587	No
Topic sentiment	T3	0.070	36	[-0.033, 0.293]	1.000	No
Topic sentiment	T4	0.135	44	[-0.153, 0.379]	0.739	No
Topic sentiment	T5	0.066	35	[-0.117, 0.246]	1.000	No
Democratic stance	T0	--	1	--	--	No
Democratic stance	T1	--	2	--	--	No
Democratic stance	T2	0.191	30	[0.000, 0.428]	0.608	No
Democratic stance	T3	0.000	6	[-0.270, 0.343]	1.000	No
Democratic stance	T4	0.049	16	[-0.228, 0.231]	1.000	No
Democratic stance	T5	0.000	5	[-0.478, 0.559]	1.000	No
Republican stance	T0	--	3	--	--	No
Republican stance	T1	--	4	--	--	No
Republican stance	T2	0.306	46	[-0.132, 0.583]	0.587	No
Republican stance	T3	0.302	21	[-0.095, 0.740]	0.587	No
Republican stance	T4	0.313	23	[-0.019, 0.534]	0.587	No
Republican stance	T5	0.312	19	[-0.173, 0.710]	0.587	No

No primary correlation survives the 36-test Benjamini-Hochberg correction. The largest positive coefficients occur for Republican stance in T4 (r=0.313), T5 (r=0.312), T2 (r=0.306), and T3 (r=0.302), but every adjusted q-value is 0.587 or larger and every block-bootstrap interval crosses zero. T2 topic sentiment has r=0.170 with N=61; its positive bootstrap interval does not translate into multiplicity-adjusted significance (q=0.587). The Democratic Sweep results therefore do not establish a statistically significant relationship during the study window.

6. Democratic Sweep Robustness and Prediction

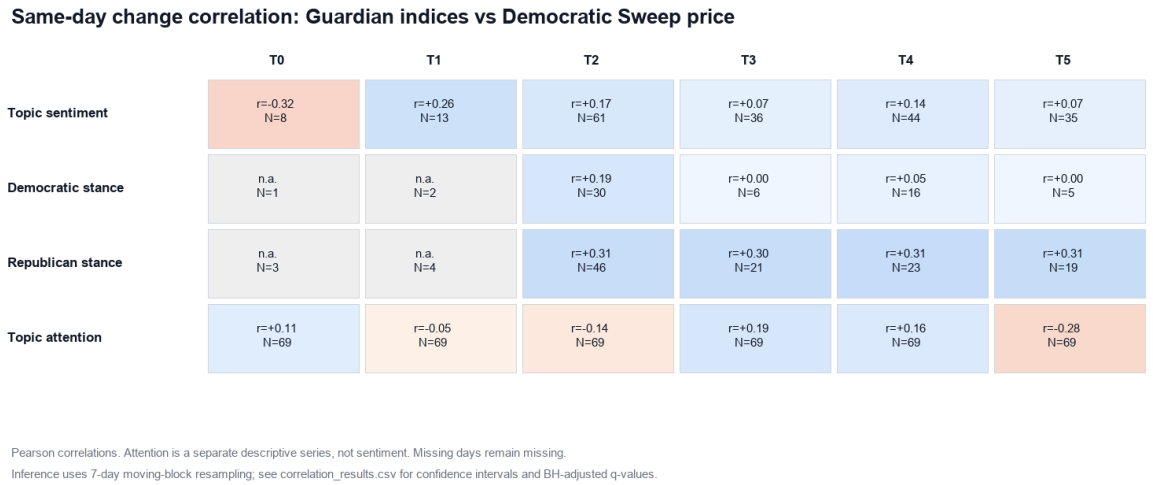


Figure 5. Primary same-day change correlations. Confidence intervals use a seven-day moving-block bootstrap that preserves calendar order and missingness. None of the estimates survives BH adjustment.

The expanding-window backtest uses S observed at the end of day t to predict the next-day market change dP. Estimation begins only after 30 historical feature-target pairs, missing feature days are skipped, and standardization parameters are estimated from the then-available training window. Two benchmarks are evaluated on exactly the same forecast dates: zero change and the historical training-window mean change.

Table 6. Walk-forward models that beat both RMSE benchmarks

Index	Topic	Feature	Pred. N	Model	Zero	Mean	R2 vs 0	R2 vs mean
Democratic stance	T2	level	18	0.01310	0.01374	0.01329	0.091	0.028
Republican stance	T2	change	16	0.01554	0.01581	0.01568	0.034	0.018
Topic sentiment	T4	level	32	0.01536	0.01551	0.01540	0.020	0.006
Topic sentiment	T2	change	31	0.01331	0.01344	0.01335	0.019	0.005

Prediction result Four of 15 evaluable specifications beat both RMSE benchmarks. The best is T2 Democratic stance in levels (18 forecasts; OOS R2=0.091 versus zero and 0.028 versus the training mean). The results indicate modest predictive content in selected policy-news signals during the study window.

7. Individual-Stock Associations

For each stock and each sufficiently observed topic index, the project estimates a univariate HAC regression of market-adjusted open-to-close return on the pre-open policy signal. Effects are reported in basis points for a one-standard-deviation increase in the signal. The main family contains 9,976 valid tests after enforcing a fixed minimum of 30 observations. Benjamini-Hochberg correction is applied globally across this family; a stricter subset additionally requires a five-day moving-block bootstrap interval to exclude zero in the HAC direction.

Table 7. Significant stock-index tests by signal

Topic	Index	BH-FDR q<.05	+ block bootstrap
T4	Democratic stance	147	62
T2	Democratic stance	120	21
T3	Topic sentiment	24	24
T5	Topic sentiment	19	19
T4	Topic sentiment	11	10
T3	Republican stance	7	7
T4	Republican stance	6	6
T2	Republican stance	4	4
T2	Topic sentiment	4	4
T5	Republican stance	4	4

Across 998 testable stocks, 346 stock-index rows (299 unique stocks) pass global FDR, and 161 rows (148 unique stocks) also pass the block-bootstrap screen. Discoveries are concentrated in T4 Democratic stance (147 FDR rows) and T2 Democratic stance (120), followed by T3 topic sentiment (24) and T5 topic sentiment (19). Concentration in a few sparse news signals increases the importance of robustness and economic interpretation.

Table 8. Representative strict stock findings

Ticker	Topic	Index	N	bps / 1SD	BH q	Bootstrap
ESS	T4	Dem. stance	31	-50.8	8.52e-09	Yes
LNG	T4	Dem. stance	31	-61.5	2.23e-07	Yes
M	T4	Dem. stance	31	+78.2	4.02e-07	Yes
TXN	T4	Dem. stance	31	-71.6	9.74e-07	Yes
CMG	T4	Dem. stance	31	-122.6	1.84e-06	Yes
ROK	T2	Dem. stance	39	-48.3	1.98e-06	Yes
TTC	T4	Dem. stance	31	+41.2	3.45e-06	Yes
PCG	T2	Dem. stance	39	+61.0	4.26e-06	Yes
CASY	T4	Dem. stance	31	-94.0	5.06e-06	Yes
CAT	T2	Dem. stance	39	-69.7	5.35e-06	Yes
WMB	T2	Dem. stance	39	-41.4	5.88e-06	Yes
POOL	T4	Dem. stance	31	+81.8	9.35e-06	Yes

8. Equity Findings: Magnitude and Interpretation

Representative strict stock effects

Market-adjusted basis points per one-standard-deviation signal increase

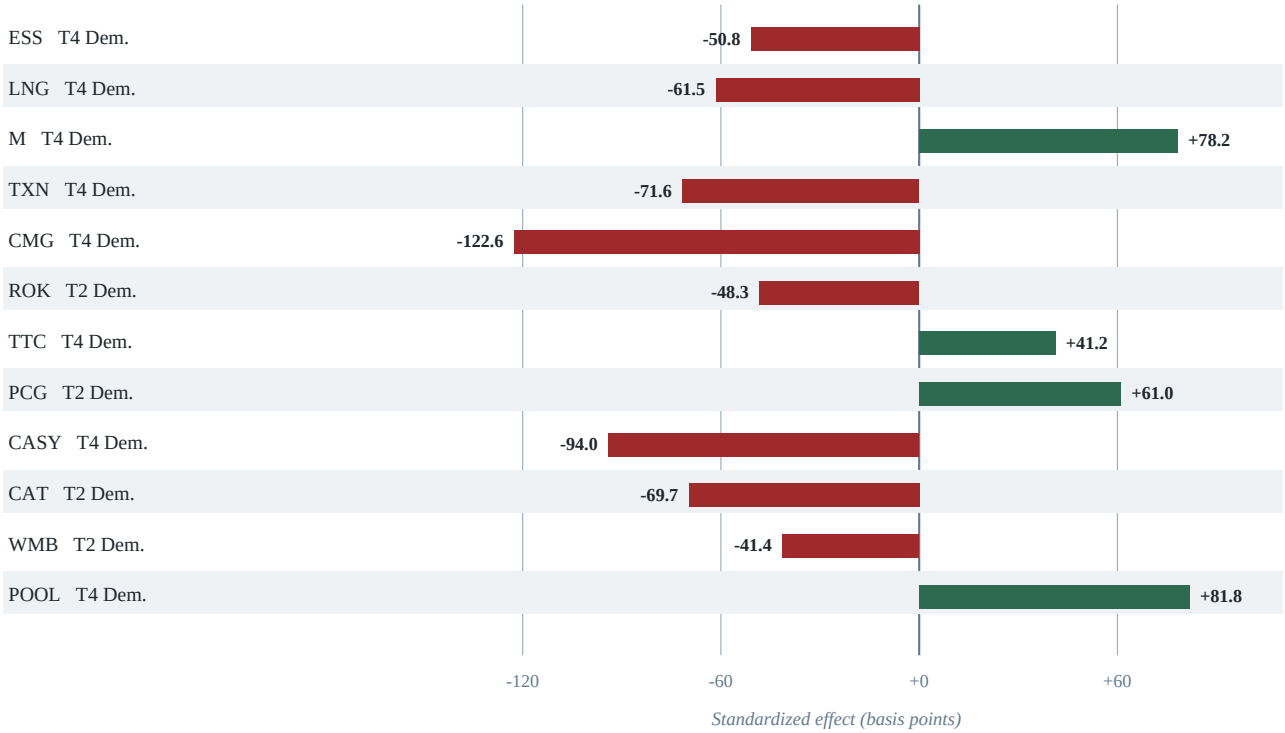


Figure 6. The same representative FDR-and-bootstrap-supported stock findings listed in Table 8. Positive and negative bars indicate the sign of the within-sample association; they are not long/short recommendations or causal effects.

Representative strict findings include negative T4 Democratic-stance associations for Essential Properties Realty Trust (ESS), Cheniere Energy (LNG), Texas Instruments (TXN), and Chipotle (CMG), alongside positive associations for Macy's (M), The Toro Company (TTC), and Pool Corporation (POOL). T2 Democratic stance is negatively associated with Rockwell Automation (ROK) and Caterpillar (CAT) but positively associated with PG&E; (PCG). These signs describe within-sample co-movement with a text-derived signal; they do not imply that a party stance caused the return.

Statistical significance	Global BH-FDR controls the expected false-discovery proportion across the tested stock-index family, while the moving-block bootstrap evaluates stability under time dependence. Together, 346 rows pass FDR and 161 also pass the bootstrap screen, showing that a substantial subset of the equity relationships is statistically robust.

9. Dedicated Midterm Baskets: Long-Only Results

Dedicated baskets pool stocks associated with the same topic-index signal. Long-only baskets equally weight constituent returns. FDR baskets use stocks with globally adjusted q-values below 0.05, while the stricter bootstrap baskets retain only relationships whose moving-block confidence intervals exclude zero in the same direction. This construction summarizes the common market response across stocks linked to each policy signal.

Table 9. All FDR-significant dedicated long-only baskets

Selection	Signal	Stocks	N	bps / 1SD	BH q	Basket bootstrap
Strict signal	T2 Dem. stance	21	39	-29.1	5.31e-13	Yes
Strict top 20	T2 Dem. stance	20	39	-28.9	9.77e-12	Yes
Strict signal	T3 Rep. stance	7	38	+67.7	1.17e-10	Yes
FDR signal	T4 Dem. stance	147	31	-29.0	1.07e-08	No
FDR signal	T2 Dem. stance	120	39	-12.7	4.88e-07	No
Strict top 20	T3 Sentiment	20	50	+26.5	9.75e-07	Yes
Strict signal	T2 Rep. stance	4	51	-46.2	2.77e-06	Yes
Strict signal	T3 Sentiment	24	50	+23.9	2.77e-06	Yes
Strict signal	T4 Sentiment	10	53	-21.3	1.61e-05	Yes
Strict top 20	T4 Dem. stance	20	31	-13.3	6.07e-05	Yes
FDR signal	T4 Sentiment	11	53	-15.9	3.28e-04	Yes
Strict signal	T4 Dem. stance	62	31	-10.8	0.023	Yes
Strict signal	T2 Sentiment	4	59	+23.5	0.040	Yes

There are 13 significant dedicated long-only baskets. The strongest are the T2 Democratic-stance strict baskets: 21-stock and top-20 versions return approximately -29 basis points per one-standard-deviation increase in the index (q values below 1e-11). T3 Republican stance and T3 sentiment long-only baskets are positive, while T4 sentiment baskets are negative. These collective patterns indicate economically meaningful topic-specific relationships in market-adjusted stock returns.

10. Dedicated Midterm Baskets: Direction-Aligned Results

Direction-aligned baskets multiply each constituent by the sign of its stock-level coefficient before equal weighting. This creates a long-short statistical factor that measures whether individual-stock responses aggregate coherently at the topic level. Reported effects are gross statistical sensitivities and do not incorporate borrow costs, turnover, spreads, or capacity.

Table 10. All FDR-significant dedicated direction-aligned baskets

Selection	Signal	Stocks	N	bps / 1SD	BH q	Basket bootstrap
Strict signal	T2 Dem. stance	21	39	+46.5	3.08e-21	Yes
Strict top 20	T2 Dem. stance	20	39	+47.1	1.25e-20	Yes
Strict signal	T4 Dem. stance	62	31	+57.4	3.83e-20	Yes
Strict top 20	T4 Dem. stance	20	31	+66.8	3.83e-20	Yes
Strict signal	T4 Sentiment	10	53	+52.2	6.89e-19	Yes
Strict top 20	T3 Sentiment	20	50	+47.7	2.16e-17	Yes
Strict signal	T3 Sentiment	24	50	+45.2	2.92e-17	Yes
Strict signal	T5 Sentiment	19	50	+57.2	6.41e-16	Yes
Strict signal	T3 Rep. stance	7	38	+78.6	4.44e-13	Yes
Strict signal	T4 Rep. stance	6	43	+69.8	6.66e-11	Yes
Strict signal	T5 Rep. stance	4	35	+103.6	1.07e-08	Yes
Strict signal	T2 Sentiment	4	59	+60.8	6.94e-08	Yes
Strict signal	T2 Rep. stance	4	51	+46.2	2.77e-06	Yes

All 13 direction-aligned dedicated baskets pass both basket-level FDR and block-bootstrap criteria. Estimated magnitudes range from +45 to +104 basis points per signal standard deviation. The results show strong cross-stock coherence and economically meaningful aggregate sensitivity to the corresponding policy-news indices.

11. Conclusions, Limitations, and Reproducibility

Table 11. Evidence summary

Component	Bottom line	Evidence
Topic modeling	Useful descriptive structure	K=6 has the best candidate coherence (0.407), low keyword overlap, and interpretable policy clusters.
Time series	Use pooled 7-day displays	They are more continuous than raw daily sentiment while retaining genuine missingness and sample-size warnings.
Democratic Sweep	No reliable association	No primary dS-dP correlation survives BH correction; OOS gains are small and sparse.
Individual stocks	Many significant associations	346 FDR rows / 299 stocks; 161 rows / 148 stocks also pass block bootstrap.
Dedicated baskets	Strong statistical associations	26 signal-specific baskets are FDR-significant; 24 also pass the basket-level block-bootstrap criterion.

11.1 Main limitations

First, the news sample is produced by a single publisher and cannot represent the electorate. Second, policy selection and topic labels involve judgment; T2 and T4 remain broad composites. Third, sentiment labels lack an independent human gold standard, and party-stance labels are asymmetric. Fourth, daily sentiment is sparse, especially for T0 and T1. Fifth, Polymarket liquidity, microstructure, and price formation are not modeled. Sixth, equity tests are observational, so the estimates describe statistical relationships rather than causal effects.

11.2 Reproducibility checks

Saved validation reports confirm: article IDs align across raw text, LDA probabilities, and annotations; all 1,356 article-topic keys are unique; topic probabilities sum to one within numerical tolerance; saved LDA, dictionary, and phrase models reload successfully; daily calendars contain all 92 dates; market timestamps are unique and in range; differencing occurs before complete-case filtering; walk-forward targets are causally available; and manual recomputations match one daily sentiment value, one weekly value, one correlation, and one forecast.

Selected project sources

The Guardian Open Platform (news corpus); spaCy and Gensim (text preprocessing and LDA); DeepSeek API (structured article annotation); Polymarket Gamma and CLOB APIs (contract verification and hourly price history); yfinance (equity and SPY price data). All numerical results in this report are read from the final validated project outputs under *data/lda_policy*, *data/policy_sentiment*, *data/sweep_analysis*, and *data/equity_sentiment*.
Project repository: https://github.com/kc6534-JerryCai/NLP_asignment5